

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – Fitness Buddy (FitVerse)

Domain of Project – Healthcare

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Problem Statement -Fitness Buddy

The Challenge:-

In today's fast-paced world, maintaining a healthy lifestyle has become increasingly difficult due to busy schedules, lack of personalized guidance, and inconsistent motivation. Many people struggle to follow effective workout routines, make healthy food choices, and build sustainable fitness habits. Traditional fitness solutions often rely on expensive gym memberships, personal trainers, or rigid programs that fail to adapt to individual goals, fitness levels, and daily routines.

As a result, there is a growing need for an intelligent, accessible, and user-friendly virtual fitness assistant that provides personalized guidance anytime and anywhere. Such a solution should offer real-time workout recommendations, nutrition advice, and continuous motivation while adapting to each user's unique needs and preferences.

FitVerse addresses this challenge by providing an AI-powered conversational fitness and wellness coach that can:

- Generate personalized home workout plans based on user goals, fitness level, and available time.
- Provide daily motivation, fitness tips, and habit-building guidance to encourage long-term consistency.
- Recommend simple, nutritious meal ideas and healthy lifestyle suggestions.
- Answer fitness-related questions with safe, practical, and easy-to-understand advice.
- Promote healthier living through personalized, AI-driven coaching available 24/7.

Proposed Solution: AI-Powered Fitness Agent

The proposed solution is **FitVerse**, an AI-powered Fitness and Wellness Agent built using **IBM watsonx.ai Granite Models**, **Python Flask**, and **IBM Cloud Lite Services** to provide personalized fitness guidance and healthy lifestyle recommendations. The platform offers an intelligent conversational interface that delivers real-time workout plans, nutrition advice, and wellness support based on user goals and preferences.

The solution consists of four key AI agents:

- **Workout Planning Agent** generates personalized home or gym workout routines based on fitness goals, experience level, available time, and equipment.
- **Nutrition Guidance Agent** recommends balanced meal plans, calorie intake, hydration goals, and healthy food suggestions tailored to user preferences.
- **Motivation & Habit Agent** provides daily motivation, fitness challenges, and habit-building tips to encourage long-term consistency.
- **Health Advisory Agent** offers safe exercise guidance, recovery tips, stretching routines, and general wellness recommendations with appropriate medical disclaimers.

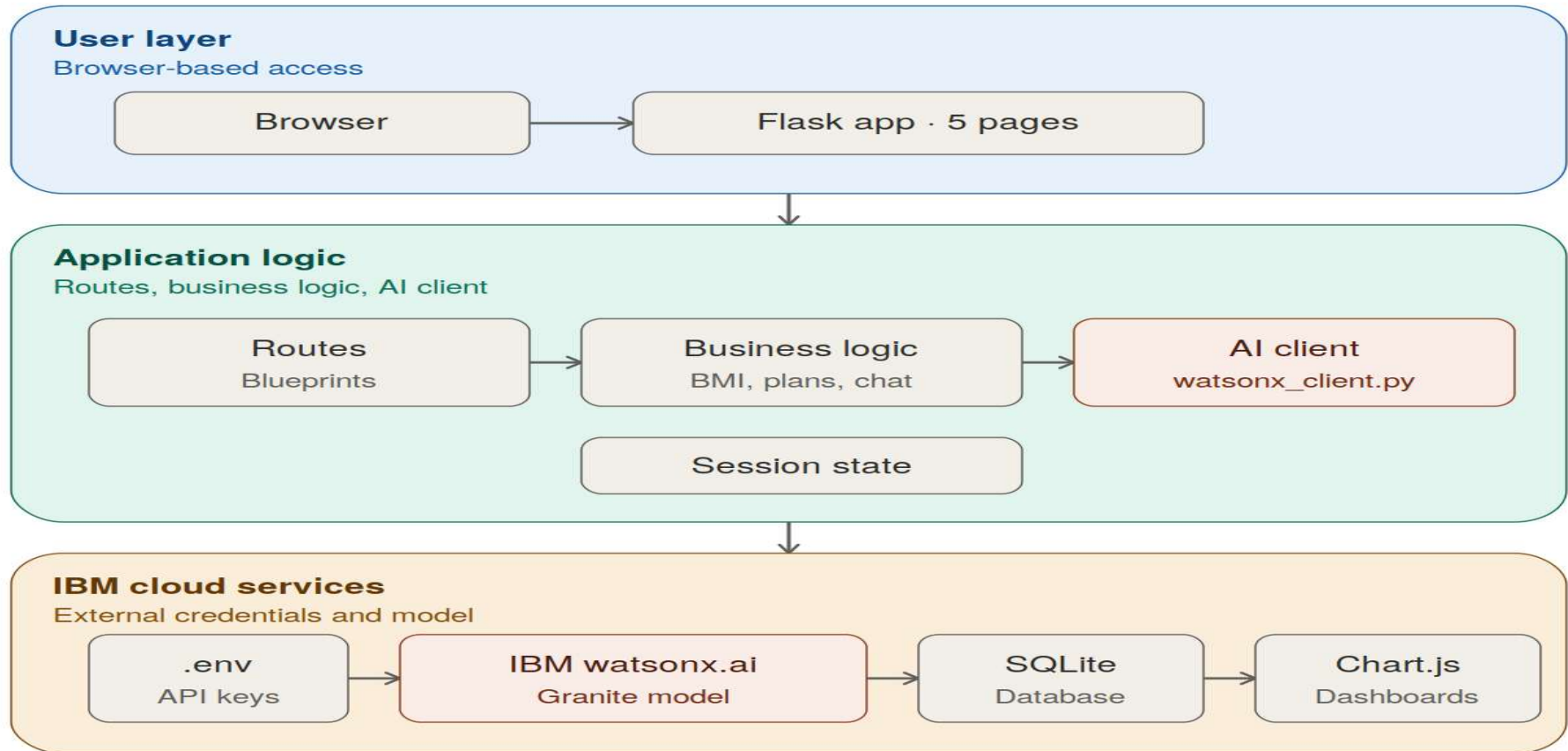
The platform also includes interactive tools such as **BMI Calculator**, **BMR Calculator**, **Calorie Calculator**, and **Water Intake Calculator**, along with a responsive dashboard that displays AI-generated workout plans, nutrition insights, health metrics, and daily fitness recommendations.

Powered by **IBM watsonx.ai Granite Models**, FitVerse delivers an intelligent, personalized, and accessible fitness coaching experience, helping users build healthier habits and achieve their long-term wellness goals.

Technology Used

- **IBM watsonx.ai Granite Model** – Powers conversational AI, personalized workout generation, nutrition guidance, and fitness recommendations.
- **IBM Cloud Lite Services** – Provides a secure, reliable, and scalable cloud platform for deploying the application.
- **Python Flask** – Backend framework for application logic, AI integration, and API management.
- **Bootstrap 5** – Builds a modern, responsive, and user-friendly interface.
- **HTML, CSS & JavaScript** – Create an interactive and dynamic frontend experience.
- **Chart.js** – Visualizes BMI, calorie estimates, hydration goals, and fitness insights using interactive charts.
- **Python Dotenv (.env)** – Securely manages IBM Cloud API keys and environment variables.

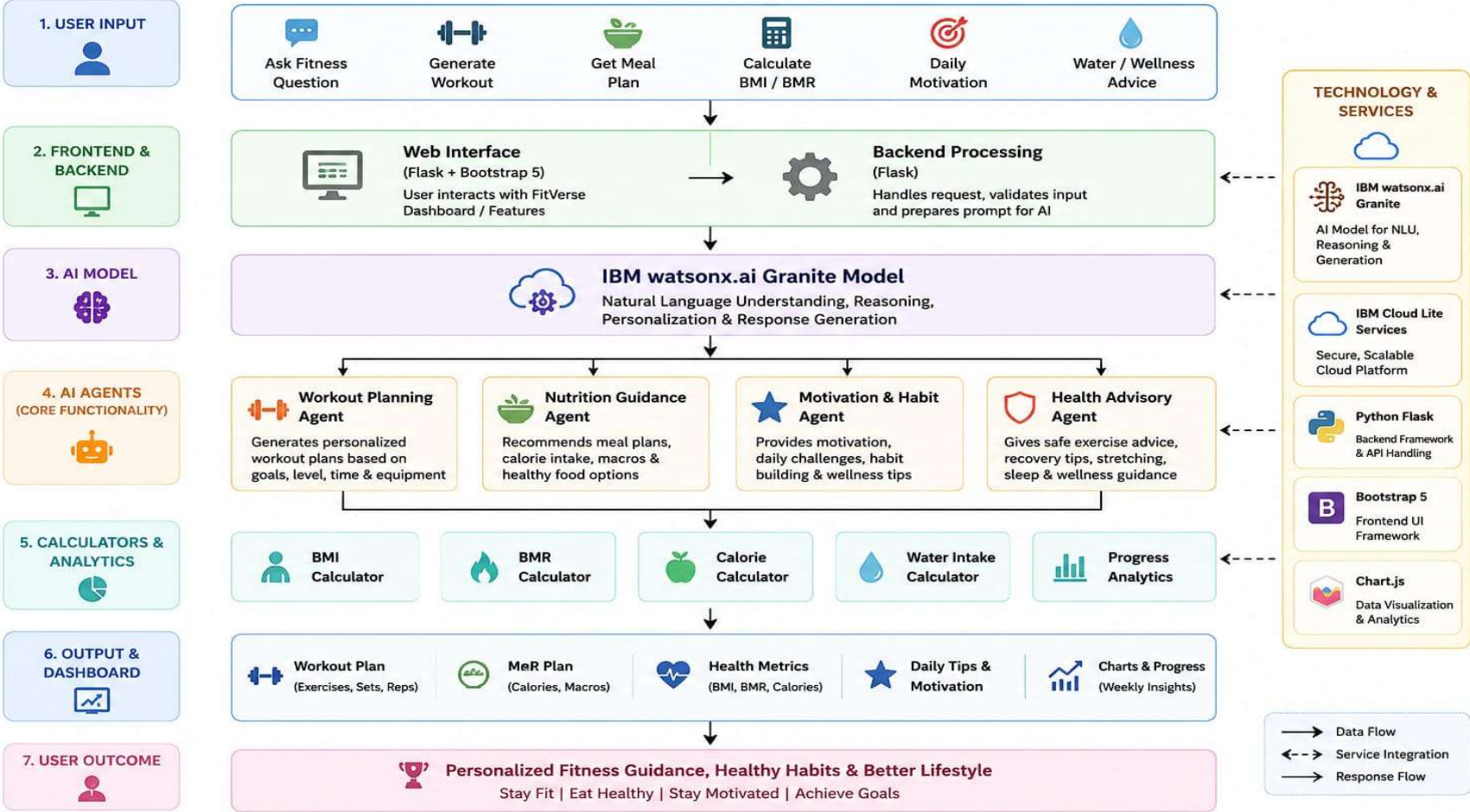
Architecture -



Architecture/Workflow Blueprint

FitVerse – AI Fitness Coach

Workflow Diagram



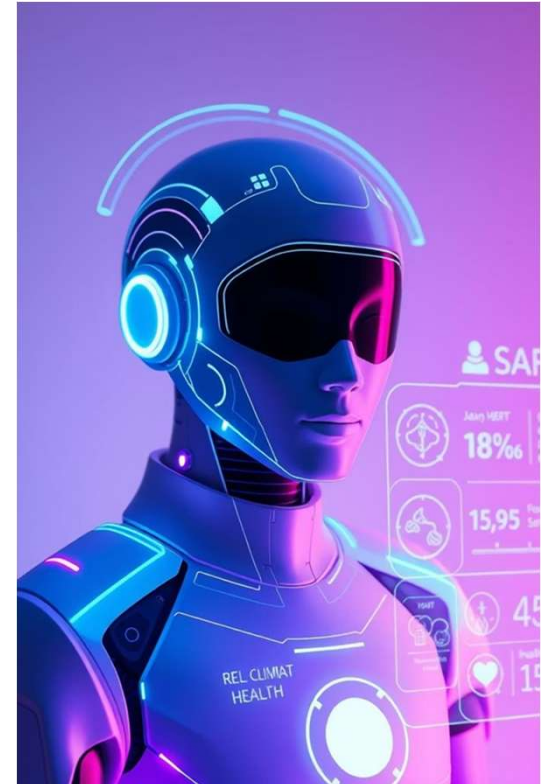
Role of Agentic AI in the solution

Role of Agentic AI in Fitness Agent:-

FitVerse leverages **Agentic AI** to function as an intelligent virtual fitness coach that understands user goals, reasons about their requirements, and delivers personalized recommendations. Instead of providing static responses, the AI analyzes user inputs such as fitness goals, workout preferences, dietary choices, and available time to generate customized workout plans, nutrition guidance, and motivational support. The Agentic AI autonomously performs specialized tasks through multiple AI agents, including:

- **Workout Planning Agent** – Creates personalized exercise routines.
- **Nutrition Guidance Agent** – Recommends healthy meals and calorie intake.
- **Motivation & Habit Agent** – Encourages consistency through daily tips and fitness challenges.
- **Health Advisory Agent** – Provides safe workout practices and general wellness recommendations.

By combining natural language understanding with intelligent reasoning using **IBM watsonx.ai Granite Models**, Agentic AI enables FitVerse to deliver real-time, context-aware, and personalized fitness coaching, making healthy living more accessible, interactive, and engaging.



Novelty and Uniqueness

The proposed **FitVerse** stands out by combining **Agentic AI**, **IBM watsonx.ai Granite Models**, and personalized fitness coaching into a single intelligent wellness platform.

- **Multi-Agent Intelligence** – Specialized AI agents collaborate to provide personalized workout plans, nutrition guidance, motivation, and wellness recommendations.
- **Personalized Fitness Coaching** – Generates customized workouts and meal suggestions based on user goals, fitness level, available time, and dietary preferences.
- **Conversational AI Experience** – Enables users to interact naturally with an AI fitness coach for instant guidance and real-time recommendations.
- **Integrated Health Tools** – Combines BMI, BMR, calorie, and hydration calculators with AI-powered insights in one platform.
- **Motivation & Habit Building** – Encourages long-term consistency through daily fitness tips, wellness challenges, and healthy habit recommendations.
- **Accessible & User-Friendly** – No login or subscription required, allowing users to access AI-powered fitness guidance instantly from any device.
- **Scalable & AI-Powered** – Built on **IBM watsonx.ai Granite Models** and **IBM Cloud Lite Services**, ensuring intelligent, reliable, and scalable performance.

FitVerse

Dashboard

AI Coach

Workout Planner

Nutrition

BMI & BMR

Habit Tracker

Profile

K

Kevin Tailor

Muscle Gain

Dashboard

BMI 27.2

Good Morning, Kevin Tailor!

Chat with AI Coach

340

CALORIES TODAY

Target: 3378.9 kcal

65min

WORKOUT TODAY

This week: 1 sessions

1.6L

WATER INTAKE

Target: 3.2L

4/4

HABITS DONE

100% complete

Weekly Workout Activity

Log Workout

Your Profile

Edit

27.2

Overweight

Body Mass Index

Weight

91.0 kg

Height

183.0 cm

TDEE

3378.9 kcal

Goal

Muscle Gain

FitVerse

Dashboard

AI Coach

Workout Planner

Nutrition

BMI & BMR

Habit Tracker

Profile

Kevin Tailor
Muscle Gain

AI Fitness Coach

BMI 27.2

FitBot — AI Coach
IBM Granite • Online

Clear

Workout Plan

Meal Plan

Calorie Target

Yoga Routine

HIIT Workout

Recovery Tips

Hydration Guide

Indian Diet Tips

give me full plan

You • 2026-07-06 11:33:52.641365

Workout Plan

As a 20-year-old male with a BMI of 27.2 and a goal of muscle gain, I've created a personalized workout plan for you, Kevin. Since you're very active and have access to a full gym, we'll focus on a 4-day split routine with progressive overload.

Day 1: Chest and Triceps

1. Barbell Bench Press (3 sets of 8-12 reps)

2. Incline Dumbbell Press (3 sets of 10-15 reps)

3. Cable Fly (3 sets of 12-15 reps)

4. Tricep Pushdown (3 sets of 10-12 reps)

5. Tricep Dips (3 sets of 12-15 reps)

Ask me anything about fitness, nutrition, workouts...

Press Enter to send · Shift+Enter for new line

0/2000

Workout Planner

AI Workout Generator

AI Ready

Fitness Goal

Muscle Gain

Fitness Level

Beginner

Intermediate

Advanced

Days/Week

5 days

Session Duration

45 min

Available Equipment

Bodyweight Only

Generate AI Workout Plan

Log Today's Workout

Weight Loss

Duration (min)

Calories burned

Recent Workouts

DATE	TYPE	DURATION	CALORIES	NOTES
Jul 07	Muscle Gain	65 min	624 kcal	—

30-DAY WORKOUT HISTORY

Date	Calories Burned (kcal)
Jun 08	0
Jun 09	0
Jun 10	0
Jun 11	0
Jun 12	0
Jun 13	0
Jun 14	0
Jun 15	0
Jun 16	0
Jun 17	0
Jun 18	0
Jun 19	0
Jun 20	0
Jun 21	0
Jun 22	0
Jun 23	0
Jun 24	0
Jun 25	0
Jun 26	0
Jun 27	0
Jun 28	0
Jun 29	0
Jun 30	0
Jul 01	0
Jul 02	0
Jul 03	0
Jul 04	0
Jul 05	0
Jul 06	0
Jul 07	624

Nutrition Planner

BMI 27.2

Calories

340

/ 3378.9 kcal

Protein

120.0g

Target: 163g

Carbs

79.0g

Daily intake

Fat

45.0g

Daily intake

AI Meal Plan Generator

AI Ready

Diet Type

Vegetarian

Fitness Goal

Weight Loss

Daily Calories Target

3378.9

Plan Duration

7 Days

Generate AI Meal Plan

Today's Meals

MEAL	FOOD	CAL	P	C	F	
Breakfast	Oats with fruits	90	30.0g	29.0g	15.0g	
Lunch	Palak paneer with rice	250	90.0g	50.0g	30.0g	

Healthy Indian Meal Ideas

BREAKFAST	LUNCH	DINNER	SNACK
Poha with peanuts	Dal rice with sabzi	Khichdi with papad	Roasted chana
Idli with sambar	Rajma chawal	Dal tadka with roti	Makhana
Upma	Chole with roti	Stir-fried vegetables with chapati	Sprouts chaat
Paratha with curd	Mixed veg with phulka	Paneer tikka with salad	Fruit with peanut butter
Oats with fruits	Palak paneer with rice		Vegetable soup
Moong dal cheela			Buttermilk

Habit Tracker

1 Day Streak!

Keep the momentum going!

"Consistency is the key to lasting change."

Today's Habits

Quick pick

Add

Or type a custom habit...

+

✓ Morning Workout

x

✓ Eat Healthy Breakfast

x

✓ 10 Minute Meditation

x

✓ Read / Learn Something New

x

Water Intake

Reset

1.6L

of 3.2L daily target

+150ml

+250ml

+350ml

+500ml

Sleep Tracker

8.25h

Last Night's Sleep

Excellent

Hours Slept

7.5

Quality

Good

Log Sleep

7-Day Overview

HABIT COMPLETION %

100

80

60

40

20

0

Wed

Thu

Fri

Sat

Sun

Mon

Tue

WATER INTAKE (ML)

1,600

1,200

800

400

0

Wed

Thu

Fri

Sat

Sun

Mon

Tue

SLEEP HOURS

10

8

6

4

2

Wed

Thu

Fri

Sat

Sun

Mon

Tue

GitHub Link + Live Demo:-

- **GitHub Link**
 - <https://github.com/KevinTailor102/Fit-Verse.git>
- **Live Server**
 - <https://fit-verse.onrender.com>

Future Scope

1. Integration with Wearables & Smart Health Devices

- FitVerse can be integrated with smartwatches and fitness trackers (such as Fitbit, Apple Watch, or Google Fit) to collect real-time data including heart rate, step count, calories burned, sleep patterns, and physical activity. This will enable the AI to generate more accurate workout plans, nutrition recommendations, and wellness insights based on live health data.

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2. AI Voice Coach & Multilingual Support

- Future versions can include a voice-enabled AI fitness coach with multilingual support, allowing users to interact naturally through voice commands in their preferred language. This will improve accessibility, enhance user engagement, and make FitVerse more inclusive for users from diverse linguistic and cultural backgrounds.

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Certificates Earned:

Credly certificate:-



Certificates Earned:

IBM BOB Certificate:-

IBM SkillsBuild	Completion Certificate
	This certificate is presented to Kevin Tailor for the completion of Lab: Troubleshoot Your Code Using IBM Bob (ALM-COURSE_4071307) According to the Adobe Learning Manager system of record
Completion date: 29 Jun 2026 (GMT)	Learning hours: 30 mins

Thank You!

Thank you for your time and interest.